

We're back at analysing zero dynamics attacks.

We've seen the delta system. While the first equation (x_{k+1}^a) is important for the damage, the second equation (y_k^a) is important for detectability.

We also need to have the initial condition given by $x_{k_0}^a = 0$, as the initial impact will be present at the next step, $k_0 + 1$.

Let's now define the attack sequence, from an initial attack time k_0 up to a final attack time k_f , given by:

$$\mathcal{A}_{k_0}^{k_f} = \{a_{k_0}, \dots, a_{k_f}\}$$

Then, we're interested in finding 0-stealthy attack sequence. $\mathcal{A}_{k_0}^{k_f}$ is 0-stealthy

$\iff y_k^a = 0 \ \forall k \geq k_0$. We can also define α -stealthy attack sequences, such that $\mathcal{A}_{k_0}^{k_f}$ is α -stealthy $\iff \|y_k^a\| \leq \alpha \ \forall k \geq k_0$.

First, we present a perfect attack case, having a 0-stealthy attack sequence, and from it we'll construct the more feasible attack, built with an α -stealthy attack sequence.

Of course we'll focus on non-trivial attacks, such that $\mathcal{A} \neq 0$.

Let's say we find $\mathcal{A}_{k_0}^{k_f}$ such that $y_k^a = 0$. This means that the attack is undetectable, and so no consistent detector can see the attack. We have that

$$y_k = \bar{y}_k + \cancel{y_k^a} = \bar{y}_k.$$

While $u_k = \mathcal{F}(y_k) = \mathcal{F}(\bar{y}_k) = \bar{u}_k$ (from above). So, if $\mathcal{A}$ is 0-stealthy ($y_k^a \equiv 0$), then $(u_k, y_k) = (\bar{u}_k, \bar{y}_k)$, and since the detector is consistent he will say no attack.

Let's now see the attack construction.

$$\begin{cases} x_{k+1}^a = Ax_k^a + Bu_k^a + Ba_k \\ y_k^a = Cx_k^a \end{cases} \quad x_{k_0}^a = 0$$

Can we find $a_k \neq 0$ such that $y^a \equiv 0$?

Imagine that a_k is a vector that has the same dimension of u_k

What is a first condition to be satisfied? We need to have $x_k^a \in \text{Ker}(C) \ \forall k \geq k_0$.

In k_0 , is $x_{k_0}^a$ in the $\text{Ker}(C)$? Yes, as $x_{k_0}^a = 0$. So, the key goal for the design of the attack is to select a_k such that x_k^a remains forever inside the $\text{Ker}(C)$.

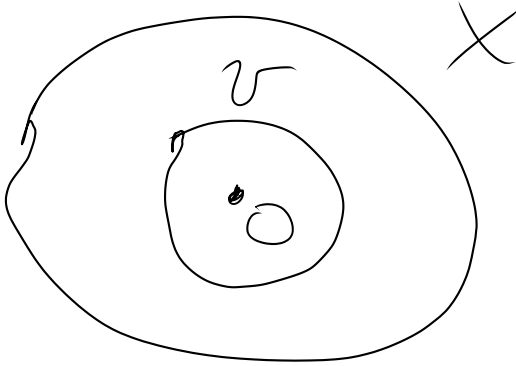
When we talk about "remains forever" in System theory it can be translated to invariance.

Consider a system:

$$x_{k+1} = f(x_k)$$

And consider the set of all variables.

The invariance is referred to a region of the space containing the origin.



A subspace $\mathcal{V} \subseteq X$ (X is the state space) is invariant with respect to $x_{k+1} = f(x_k)$
 $\iff \forall x_{k_0} \in \mathcal{V}$ then $x_k \in \mathcal{V} \forall k \geq k_0$.

Why is invariance important? If a given subset $\mathcal{V}$ has an important property, and we start the system in $\mathcal{V}$, then we're sure that that property will hold forever. An important property: $x_k^a \in \text{Ker}(C)$. We thus want to have $\mathcal{V} \subseteq \text{Ker}(C)$.

We want to pass from simple invariance to another type of invariance, as $f(x)$ is an autonomous equation.

We thus define controlled invariance. We have $x_{k+1} = f(x_k, u_k)$.

We say that $\mathcal{V} \subseteq X$ is controlled invariant with respect to $x_{k+1} = f(x_k, u_k) \iff \forall x_k \in \mathcal{V}, \exists u_k : x_{k+1} = f(x_k, u_k) \in \mathcal{V}$.

Let's now see a particular case for linear system.

$\mathcal{V} \subseteq X$ is controlled positive invariant (or "(A+BF)-invariant") $\iff \forall k, \forall x_k \in \mathcal{V}, \exists u_k = F \cdot x_k$ (where F is the feedback matrix) such that

$$x_{k+1} = Ax_k + Bu_k = (A + BF)x_k \in \mathcal{V}.$$

Note: $u_k = F \cdot x_k$ makes the new system $x_{k+1} = (A + BF)x_k$ invariant.

For linear systems, we say:

- $\mathcal{V}$ is A -invariant (simple invariance).
- $\mathcal{V}$ is $(A + BF)$ -invariant (controlled invariance).

We now want to find a value of a_k to make a given set $\mathcal{V} \subseteq \text{Ker}(C)$ to be controlled invariant. From the state space equations, we can remove u_k^a , as if $y_k^a \equiv 0 \rightarrow u_k^a \equiv 0$.

Let's see the attack formulation. The attack block is a dynamical system, specifically a copy of the system under a linear feedback law (with matrix F). We call z_k the state of the attack system. We can define:

$$\begin{aligned} z_{k+1} &= Az_k + BFz_k = (A + BF)z_k \\ a_k &= Fz_k \end{aligned}$$

What we inject as the value in the control communication channel is Fz_k . If, however, $Fz_k \equiv Fx_f$, we're doing invariance under control, and the attack matrix F is selected to make the system controlled invariant.

We need to design F . Attacker must select F and z_{k_0} .

The attack system is a copy of the system under a linear feedback.

Now, F is chosen to make the attack system controlled invariant with respect to a given $\mathcal{V} \subseteq \text{Ker}(C) \subseteq X$, and also maximising $\mathcal{V}$. We can call it $\mathcal{V}^*$. So, the goal of the attacker is to find F that results in largest possible $\mathcal{V}^*$ that is $(A + BF)$ -invariant. There are algorithms giving $\mathcal{V}^*$ and associated F .

Let's proceed with the attack analysis. We focus on the delta system and the attacker system. We've seen how $a_k = Fz_k$, so the value of the attack is based on the state of the attack system.

Let's pretend the initial states are the same, so if $x_0^a = z_{k_0} \neq 0$? This scenario is known as matched initial conditions. We can see how the two systems are a copy, having the same equations. The attacker though cannot measure directly x_k , and so it closes the loop from z_k . If we assume that the initial states are the same, then we're basically almost closing the loop from x_k . At k_0 , we have

$$\begin{aligned} x_{k_0+1}^a &= Ax_{k_0}^a + BFz_{k_0}, \text{ with } y_{k_0}^a = Cx_{k_0}^a = 0 \text{ and } z_{k_0+1} = (A + BF)z_{k_0} \text{ and} \\ a_{k_0} &= Fz_{k_0}. \end{aligned}$$

The matched conditions state that $z_{k_0} = x_{k_0}^a \neq 0$. The two systems evolve in parallel.

In matched conditions: $x_k^a \equiv z_k \forall k \geq k_0$.

F is chosen to make $\mathcal{V}^*$ as large as possible, $(A + BF)$ -invariant and in $\text{Ker}(C)$.

Thus, we have the following result (ZDA in matched conditions):

Consider the attack system $\begin{cases} z_{k+1} = (A + BF)z_k \\ a_k = Fz_k \end{cases}$ with matched initial conditions

$z_{k_0} = x_{k_0}^a \neq 0$ (which are not true in practice), where F is chosen to make a given $\mathcal{V}^* \subseteq \text{Ker}(C) \subseteq X$ as large as possible and $(A + BF)$ -invariant. Then $z_k \equiv x_k^a \forall k \geq k_0$ and $y_k^a = Cx_k^a = Cz_k = 0$, and thus the attack is undetectable (0-stealthy).

In reality, though, $x_{k_0}^a = 0$, while $z_{k_0} \neq 0$.

We decide to study the difference between the two values, which we denote as $e_k := x_k^a - z_k$. One strategy would be to choose the attack such that we go back into the matched case after a while, leading to undetectability. We're leaving some traces though, having an initial detectable phase.

Let's study e_k . We have that $e_{k+1} = x_{k+1}^a - z_{k+1} = Ax_k^a + BFz_k - (A + BF)z_k = Ax_k^a + \cancel{BFz_k} - Az_k - \cancel{BFz_k} = A(x_k^a - z_k) = Ae_k$. We use e_k instead of x_k^a . We thus get this new system:

$$\begin{cases} e_{k+1} = Ae_k \\ z_{k+1} = (A + BF)z_k \end{cases}$$

We've written the equations of delta system but in e_k variables, plus the attack system.

We get that $y_k^a = Cx_k^a = Ce_k + \cancel{Cz_k} = Ce_k$, and the attack is $a_k = Fz_k$.

We can write this also in vector form:

$$\begin{pmatrix} e_{k+1} \\ z_{k+1} \end{pmatrix} = \begin{pmatrix} A & 0 \\ 0 & (A + BF) \end{pmatrix} \begin{pmatrix} e_k \\ z_k \end{pmatrix}$$

$$y_k^a = (C \quad 0) \begin{pmatrix} e_k \\ z_k \end{pmatrix}$$

and $z_{k_0} \neq 0$ to be specified by the attacker, while the initial condition of $e_{k_0} = -z_{k_0}$.

We now see recover of matched conditions. We focus on the $e_{k+1} = Ae_k$ and $e_{k_0} = -z_{k_0}$ equations, with z_{k_0} chosen by the attacker.

Can we (the attacker) choose z_{k_0} to make $e_k \rightarrow 0$?

We have that $e_{k_0+1} = -Az_{k_0}$, and $e_{k_0+2} = -A^2z_{k_0}$.

Let's take $k_0 = 0$, and thus we get that $e_k = -A^k z_0$, and this is the solution.

How do we choose z_0 ? Call $\lambda_1, \dots, \lambda_n$ the eigenvalues of A . Assume they're all real and distinct. Then the associated eigenvectors are n , and linearly independent, $u_1, \dots, u_n$. Remember that $Au_i = \lambda_i u_i \forall i$. This is the definition of eigenvectors/eigenvalues.

We write $z_0 = c_1 u_1 + c_2 u_2 + \dots + c_n u_n$ where c_i is the coordinate of z_0 along u_i .

Let's compute $e_k = -A^k z_0 = -A^k (c_1 u_1 + \dots + c_n u_n) = -c_1 \lambda_1^k u_1 - \dots - c_n \lambda_n^k u_n$.

Does this go to zero? Terms associated to $|\lambda_i| < 1 \rightarrow \lambda_i^k \rightarrow 0$. These are the stable eigenvalues. If $|\lambda_i| > 1 \rightarrow \lambda_i^k$ will explode.

Assume we have $e_k = -c_1 \cdot 0.5^k \cdot u_1 - c_2 \cdot 2.2^k \cdot u_2 - c_3 \cdot (-1.6)^k \cdot u_3$.

How to select z_0 ? Remember that c_1, c_2, c_3 are the coordinates of z in the new system u_1, u_2, u_3 .

We have $c_i = 0 \forall i : |\lambda_i| > 1$. Thus, z_0 is orthogonal to unstable eigenvectors.

We eventually recover matched conditions, so eventually we have that $x_k^a \rightarrow z_k$, and this will imply undetectability.